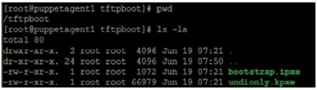

Figure 4: Razor iPXE bootstrap script

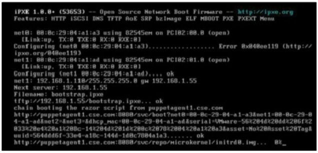
Figure 5: Razor micro-kernel booting and inventorying its facts


Figure 6: Bootloader loading the kernel image

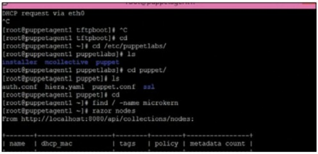

Figure 7: Status of Razor nodes before provisioning

After you've followed one of the above installation methods, you should be able to go to *http://localhost:8080/api* and get the API entry point that will give you a JSON document that talks about *collections* and *commands*.

***Setting up PXE:*** Type the following commands (see Figure 4).

```
 # wget http://links.puppetlabs.com/pe-razor-ipxe-firmare-3.3.
#  cp undionly-20140116.kpxe /var/lib/tftpboot
# cp bootstrap.ipxe /var/lib/tftpboot
```

***Configuring DNSMASQ:*** Set the following configuration under */etc/dnsmasq.conf*.

```
# This works for dnsmasq 2.45
# iPXE sets option 175, mark it for network IPXEBOOT
dhcp-match=IPXEBOOT,175
dhcp-boot=net:IPXEBOOT,bootstrap.ipxe
dhcp-boot=undionly.kpxe
# TFTP setup
enable-tftp
tftp-root=/var/lib/tftpboot
dhcp-range=192.168.1.50,192.168.1.150,12h
```

This completes the Razor server configuration. Now let's create a new VM and try to PXE boot.

As you've seen above, the new VM listened to the net1 interface and acquired the IP address from the DHCP server. Next, Razor discovers its characteristics by booting it with the Razor micro-kernel and inventorying its facts. Meanwhile, you can check the status of nodes as shown in Figure 7.

As seen in this figure, as of now, there are no provisioning elements created for the new PXE booted VM. It's time to create the provisioning elements of Razor. I have created a provisioning element for CentOS 6.5 x64.

## Creating the repository

To create a repository, type:

```
[root@puppetagent1 ~]# razor create-repo --name=CentOS6.5-Repo --iso-url http://192.168.1.100/OS/Linux/CentOS/CentOS-6.5-x86_64-bin-DVD1.iso --task centos6
From http://localhost:8080/api:
     name: CentOS6.5-Repo
     iso_url: http://192.168.1.100/OS/Linux/CentOS/CentOS-6.5-x86_64-bin-DVD1.iso
      url: ---
     task: ---
  command: http://localhost:8080/api/collections/commands/1
```

## Creating a broker

To create a broker, type:

```
[root@puppetagent1 ~]# razor create-broker --name foo --broker-type puppet-pe --configuration '{ "server": "puppetmaster.cse.com" }'
From http://localhost:8080/api:
           name: foo
          broker-type: puppet-pe
          configuration:
                   server: puppetmaster.cse.com
          policies: 0
          command: http://localhost:8080/api/collections/commands/2
```

## Creating policy

To create a policy, type: